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Part 7. On the road: your homelab, Pi-hole, and VPN away from home

> **The payoff of this part:** a clear, honest map of what you can and can't have when you leave the house, reaching your homelab remotely, carrying Pi-hole's ad-blocking with you, and running a VPN, plus the one rule that explains why you can't always have all three at once on the same device.

At home, everything is easy: your devices share the Pi's network and everything just works. The moment you walk out the door, the only thing connecting your phone to your Pi is **Tailscale**. This part is the field manual for that world.

Reaching your homelab from anywhere (the easy part)

This one already works, and it's the cleanest win. Because Tailscale is a mesh VPN, any device signed into your tailnet reaches the Pi from anywhere, coffee shop, cellular, hotel, with no extra setup:

- **Audiobookshelf:** `https://abs.home` streams just as it does at home.
- **The dashboard:** `https://home.home` (or `https://homelab`) opens from anywhere.

The reason it's effortless is that you're only reaching *into* your own network over an authenticated tunnel. The hard parts below are all about the *outbound* direction, what your traffic does and which DNS resolves it.

Pi-hole already follows you (set up in Part 4)

You don't need to do anything here, when you made Pi-hole your tailnet's DNS in Part 4 (the **Override local DNS** setup), you also got Pi-hole's ad-blocking on every device, everywhere. On cellular, your phone still resolves through Pi-hole, so ads stay blocked. The same step is also what makes your `.home` names work away from home.

That's the good news. The catch is what happens when you turn on *another* VPN, which is the rest of this chapter.

> **!** This only holds while Tailscale is the active VPN
>
> The whole mechanism rides on Tailscale being the thing controlling your device's DNS. The instant another VPN takes over (next section), this stops applying. That's not a bug, it's the central tension of going mobile, and the rest of this chapter is about navigating it.

The one rule that governs everything off-network

Here it is, the rule that explains every "why can't I just…" on the road:

> **A full-device VPN owns the default route *and* the DNS. A phone runs only one VPN at a time.**

Two consequences fall out of it:

1. **Aura and Tailscale can't both be on (on a phone).** iOS and Android allow exactly one active VPN. Turn on Aura to change location, and Tailscale drops , which means your Pi-hole-over-Tailscale push and your `https://homelab` access both go dark, because the Pi's `100.x` address is only reachable over Tailscale.
2. **Whatever VPN is active forces its own DNS.** Even setting the one-VPN limit aside, the active VPN routes DNS through its own resolvers. So while you're on *any* location/privacy VPN, your traffic is *not* going through Pi-hole, by design, to prevent DNS leaks.

`Insight` This is why "use my own VPN *and* filter through Pi-hole, on my phone, away from home" has no clean answer: Pi-hole lives only on your tailnet, reaching it needs Tailscale to be the active VPN, and a phone can't run two VPNs. The only ways out are to (a) make Pi-hole publicly reachable (a security no-go) or (b) make the VPN and Pi-hole live on the *same box you route through*, which is the Mullvad-via-Tailscale exit-node architecture, not a separate app like Aura.

The decision matrix

You can't have all three of {homelab access, Pi-hole filtering, privacy VPN} on one device at one time. But you *can* switch between coherent modes in seconds. Pick per situation:

You're away and you want…	Set this	Pi-hole?	Homelab?	Hides IP?
Homelab + ad-blocking (the everyday default)	Tailscale on, no exit node	via DNS push		(your real IP)
Appear at home / reach LAN	Tailscale + **Pi as exit node** (Part 6 C)			(home IP)

You're away and you want…	Set this	Pi-hole?	Homelab?	Hides IP?
Privacy / change location	**Mullvad exit node** (Part 6 A)	Mullvad's DNS	(LAN access on)	
Privacy via your existing app	**Aura** on	Aura's DNS	(Tailscale off)	

The pattern to read out of that table:

- **The Mullvad exit node (Option A) is the only privacy choice that keeps your homelab reachable**, because it *is* Tailscale, your tunnel to `homelab` stays up while Mullvad handles egress. It does, however, hand DNS to Mullvad, so Pi-hole is bypassed (use Mullvad's own ad-blocking DNS as a substitute).
- **Aura gives you privacy but takes everything else down** while it's on, since it displaces Tailscale entirely on the phone. Fine for a quick "look like I'm elsewhere" session; not a mode you live in if you want your homelab.
- **The Pi-as-exit-node mode is the sweet spot when you want Pi-hole filtering on the road**, your traffic goes home, gets filtered by Pi-hole, and exits from your home IP. No privacy, but full ad-blocking and full homelab access.

Making a Mullvad exit node behave on the road

If you go with the Mullvad add-on (Part 6's recommendation), two flags matter when you're mobile:

```
# Keep access to LAN devices (printers, etc.) while using an exit node
tailscale set --exit-node=<mullvad-node> --exit-node-allow-lan-access
```

- **`--exit-node-allow-lan-access`**: by default, turning on *any* exit node cuts your access to the local network you're physically on. This flag restores it. On mobile, it's the **Allow LAN access** toggle when you pick the exit node.
- **DNS leaks:** allowing LAN access can let some DNS queries escape the tunnel. Recent Tailscale clients (1.48.3+) route DNS correctly through Mullvad exit nodes without extra config, keep your clients updated.

> ⚠ A real bug to watch for
>
> Some Tailscale client versions around 1.82 had a defect where an **active exit node bypassed the global Pi-hole nameserver**, so your ad-blocking would quietly stop whenever an exit node was on, even when you expected the DNS push to apply. If you see that symptom, update the Tailscale client first. It's a known issue, not your misconfiguration.

A practical everyday routine

For most people the comfortable default is simple:

1. **Leave Tailscale on, no exit node, all the time.** This is the everyday mode: homelab reachable, Pi-hole ad-blocking everywhere via the DNS push, and your phone behaves normally. You give up IP-hiding, which most of the time you don't need.
2. **When you specifically want privacy** (sketchy Wi-Fi, want to change region): flip on the **Mullvad exit node** for that session. Accept that Pi-hole steps aside for Mullvad's DNS while it's on, then flip it back to None when you're done.
3. **Retire Aura, or keep it only as an occasional standalone tool.** Anything Aura does for you on the road, the Mullvad exit node does too, while staying inside the one VPN that keeps the rest of your homelab alive. The only reason to reach for Aura is if you specifically want its app or its kill switch on a device where you're not using Tailscale anyway.

Recap: and the whole series in one breath

- **Reaching your homelab away** is the free win: Tailscale makes `homelab:13378` and `https://home.home` work from anywhere, no extra setup.
- **Pi-hole on the road** takes one change, add the Pi as a custom nameserver in the Tailscale admin console and turn on **Override local DNS**, and works only while Tailscale is your active VPN.
- **The governing rule:** a full-device VPN owns DNS, and a phone runs one VPN at a time, so homelab access, Pi-hole filtering, and a privacy VPN can't all be on at once. Switch between modes using the decision matrix.
- **Best privacy-with-homelab combo:** the Mullvad exit node, because it lives inside Tailscale. **Best ad-blocking-on-the-road combo:** the Pi as your exit node. **Aura** stays a standalone fallback.

With that, your homelab is fully mobile: an always-on Raspberry Pi that streams your media, blocks ads on every device, presents a single `https://home.home` control panel, and (when you understand the one-VPN rule) gives you a deliberate, switchable answer for privacy and ad-blocking the moment you step out the door. All of it private by default, over one Tailscale network, with nothing exposed to the public internet.

One direction is still missing: getting your *phone and your Linux machines* to talk to each other for everyday file transfer and clipboard sharing. That's Part 8, the finale.